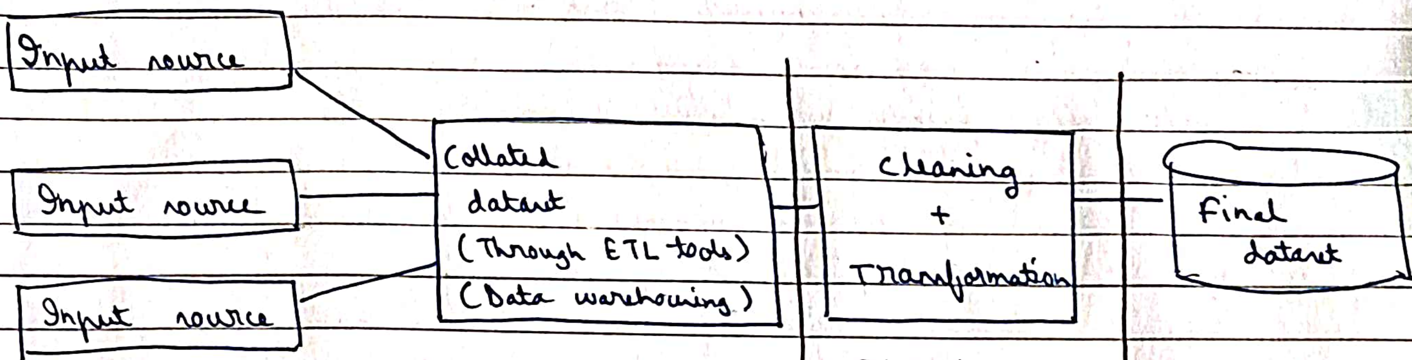
* EDA(Exploratory Data Analysis)



EDA is
the process of
converting raw
data to clean
data.
(EDA is data
cleaning techniques)

* EDA Technique == Feature engineering technique

① Variable Identification
② Univariate analysis
③ Bivariate analysis
④ outlier treatment
⑤ missing value treatment
⑥ Imputation technique (transformer)
⑦ Variable creation

① Variable Identification

Independent variable
(non target attribute)
(non labelled attribute)
(non predictive variable)
x

Dependent variable
(target attribute)
(labelled attribute)
(predictive variable)
y

Eg1: family (4 variable)

dad - employee -- dependent variable - - y

mom - housewife - independent variable (x_1)
1kid - 4th grade - " (x_2)
2kid - 2nd grade - " (x_3)

$y = x_1 + x_2 + x_3$ (multiple linear regression algorithm)

Eg 2:

family
dad (y)
mom (x)
kid (x_2)
$y = x + x_2$

Machine learning

① regression -- if dependent variable is continuous

continuous variable -- petrol price
stock price
gold price
flight price
ev car sale price

when it keeps on only increasing or decreasing

To build these projects we need to apply REGRESSION ALGORITHMS.

Simple linear regression
multiple linear regression
Gradient descent / stochastic gd / Batch gd
L1 regression / lasso regression regularization
L2 regression / ridge regression regularization
polynomial regression
support vector regressor
k nearest neighbour regression
decision tree regressor
random forest regression algo
xgboost regression (extream gradient boost)
lgbm (light gradient boosting) regression
artificial neural n/w regression
time series

② Classification : dependent variable is binary or category
win / loss (1-0)
profit / loss
happy / sad
pass / fail
yes / no

③ Clustering : no binary or continuous dependent variable

Eg:

x9	x1	x2	x3	x4	x5	x6	x7	x8	y	
gender	name	bhk	addr	school	hospital				price	Regression

x9	x1	x2	x3	x4	x5	x6	x7	x8	x10	y
gender	name								price	purchase NO Yes

relevant irrelevant variables

while building ml model please do not consider irrelevant attributes consider only relevant attributes

② univariate analysis

plot the graph using 1 variable only.

③ bivariate analysis

plot the graph using 2 variables

correlation: relation among the attribute is called corelation

0 to 1 -ve

-1 to 0 -ve

correlation range is b/w -1 to 1

④ Outlier treatment

how to detect outlier (by visualizations)

⑤ Missing value treatment

number (numerical data)

text (categorical data)

in the dataset if any value is missing then we use the following strategy

- mean
- median
- mode

Number		categorical	
10		summer	
20		winter	
$\boxed{15.25}$	mean	rainy	
23		$\boxed{\text{rainy}}$	mode
8		summer	
23		winter	
23		rainy	
		rainy	

Temp	Categorical	
45	summer	
3	winter	
23	rainy	
$\boxed{56}$	$\boxed{\text{summer}}$	KNN (nearest neighbour)
$\boxed{46}$	summer	
7	winter	We cannot take mode here
13	rainy	

if any missing values are there in the dataset :-
numerical value are missing -- mean, median, mode
categorical value are missing -- mode, KNN

⑥ Immutation technique : impute categorical data to numerical data

<u>Dummy variable or one hot encoder</u>

temp	categorical	summer	winter	rainy
46	summer	1	0	0
6	winter	0	1	0
13	rainy	0	0	1
	rainy	0	0	1
45	summer	1	0	0
7	winter	0	1	0
5	rainy	0	0	1

label encoder

categorical	
summer	0
winter	1
rainy	2
rainy	2
summer	0
winter	1
rainy	2

label encoder

categorical	
summer	0
winter	1
rainy	2
rainy	2
summer	0
winter	1
rainy	2